

Eco Plating — Browser Automations

Capability & Risk Matrix — prepared for Steelhead as input to the upcoming “*Automations*” feature.

This matrix lists every browser-level tool Eco Plating uses today on top of Steelhead. It is organized around the two dimensions we agreed to evaluate: **data-safety footprint** and **operational importance**. These tools were built as a value-add to accelerate our Steelhead adoption and to cover operational needs the platform does not yet handle natively. Our intent in sharing this is to give your team a clear road map so the native “*Automations*” feature can absorb as much of this as possible.

How these tools touch Steelhead (shared safety model)

The following holds for **every** tool below and is the basis for the per-tool data-safety rating:

- **Same session, same permissions.** Everything runs inside the signed-in user’s own browser session, with that user’s existing role and permissions. The tools do only what the same person could do by hand in the UI — no admin/elevated access.
- **Platform’s own operations.** They reuse the very same operations the Steelhead web app itself issues; they do not open any private or undocumented channel. **Your server-side validation and safeguards stay fully in effect.**
- **Conservative request volume.** Bulk work is deliberately throttled (small batches, capped concurrency) to be gentle on the platform — not a flood of uncontrolled calls.
- **Mostly reversible & auditable writes.** The large majority of writes are reversible (archive, not hard-delete) and carry an audit record (who / when / what). The few higher-impact tools are marked † below and are explicitly gated behind a user confirmation step.
- **Proactive break detection.** An automated daily check detects when Steelhead rotates its operation identifiers, so affected tools are flagged and fixed quickly — usually before a user hits an error.

† **Higher-impact operations** (each requires explicit confirmation before running): *inventory-reset* (archives existing batches + creates new ones), *process-canon* apply mode (rewrites a process node tree via the platform’s own ProcureTree), and *report-liberator*’s optional empty-folder delete.

Legend

Native in Steelhead?

No = no equivalent today.

Partial = exists but per-record / buried / no bulk path.

Data access

Read-only = never modifies data.

Pre-fill = fills a form the user reviews & saves.

Triggers native = invokes a built-in Steelhead action.

Write (rev.) = writes, reversible + audited.

Write † = higher-impact, confirmation-gated.

Importance

Critical / High / Medium / Low / One-time.

Status: Production / Live-validated / Pending validation.

Yellow rows = planned, not yet built.

A. Bulk master-data load & maintenance

Loading and maintaining part numbers, specs, processes, files and pricing at scale.

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
bulk-upload Carga Masiva	Bulk create/update of part numbers from Excel: metal, line, finishes, default process, quality specs/params, dimensions, and quotes/pricing — with an audit stamp per part.	No	Write (rev.)	Critical Production (200–4,270/run)
spec-params-bulk Carga masiva Spec Params	Export all quality-spec parameters to Excel, edit acceptance ranges / sampling / requirement flags in bulk, and upload back with a pre-change diff.	No	Write (rev.)	High Pending validation
spec-migrator Ajuste Masivo de Specs	Spec maintenance across parts: migrate to a newer spec version, assign pending params, resolve incompatible specs, and detect/clean duplicated parameters.	No	Write (rev.)	High Production
archiver Archivador de PNs	Bulk archive/unarchive part numbers by combinable criteria (created/modified/last-used date, tags with AND/OR), with a count preview before running.	No	Write (rev.)	Medium-High Live-validated
integrity-tiers Auditor de PNs	Scans all parts and detects duplicates at three severity tiers (exact quote-key, functional name+customer+metal+finishes, partial subset); bulk-archives the discards.	No	Write (rev.)	High Production (5k–12k parts)

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
process-canon † Canon de Procesos	Audits every manufacturing process against the 9-node canonical structure (in/out warehouse SPs, racking, ready-to-process scanner, drying, inspection & packing) and, in apply mode, rewrites the process tree to conform and creates the missing scanner nodes.	No	Read / Write †	Critical Production
file-uploader Cargador de Archivos	Bulk-uploads files (drawings, specs, docs) and auto-links each one to its part number by matching the file name to the part name — upload + link in one pass.	No	Write (rev.)	Medium Validated

B. In-form entry assistants (auto-fill & calculation)

Reduce keystrokes and capture errors on existing Steelhead screens. Most only *pre-fill* fields the user then reviews and saves.

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
create-order-autofill Crear OV — Autofill	On sales-order creation, fills three required custom fields: billing legal entity (from the customer's tax catalog), currency, and the "consolidate by product" flag (auto-set for specific ship-to plants).	Partial	Pre-fill	High Pending validation
proceso-calculator Calculadora de Procesos	While creating/editing a part, suggests the correct production process from the metal + line + finish-tag combination (1,580-entry catalog); can add a missing combination to the shared catalog.	No	Pre-fill	Medium Pending deploy/validation
unit-autoconvert Auto-conversión de Unidades	When editing a part's unit factors, auto-computes the equivalent units (KG↔LB, cm²↔dm²↔ft²) so price-by-weight/area factors stay consistent.	No	Pre-fill	Medium Pending validation
receiver-date-override Fecha de Recibo	Adds an editable "actual received date/time" on the customer-receiving modal, so a back-dated arrival is recorded at creation instead of edited afterward.	No	Pre-fill	High Production (issue tracked)

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
warehouse-location-prefill Ubicación de Recibo	Adds a header “initial location” on the receiving modal and propagates it to every line, instead of picking the location row by row.	Partial	Pre-fill	Medium-High Dev-validated
weight-quick-entry Peso Rápido	Adds an inline weight field (KG/LB by customer) next to the quantity field on each receiving line, so weight is captured in the same place.	Partial	Pre-fill	Medium Pending validation
sensor-status-autofill Auto-asignar status (Sensores)	On maintenance sensor panels, auto-marks the active “use for status” parameter (assists when there are several candidates, skips when none).	No	Write (rev.)	Medium Pending validation
engineering-calc Planned	Engineering helper that computes pieces per load (rack/barrel capacity) and the part’s surface area from its geometry/dimensions, feeding price-by-area and load-capacity planning on the part. <i>Scope still being defined.</i>	No	TBD (planned)	High Not yet developed
tank-autoroute Planned	When a work order is going to be produced on a <i>different production line</i> than its default, automatically re-routes the plating tanks/baths to that line’s equivalent stations and adjusts the order’s process path. <i>Scope still being defined.</i>	No	TBD (planned)	High Not yet developed

C. Billing, invoicing & tax (SAT) compliance

Accounting accuracy and Mexico-specific fiscal (SAT/CFDI) correctness on the invoicing flow.

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
cfdi-attacher Adjuntar CFDI	When sending the invoice email, automatically attaches the stamped CFDI XML (Mexican e-invoice) to each invoice in the message; if the XML can’t be attached, it blocks the send rather than mail an incomplete invoice.	No	Triggers native	Critical (fiscal) Production

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
invoice-autofill Invoice Autofill	On invoice creation, auto-fills the required accounting fields from the selected customer: AR/receivable account and the income account on each line (customer catalogs reach 60+ candidate accounts).	No	Pre-fill	Critical Production — gates legacy retirement
bill-autofill Bill Autofill	On vendor bills (AP), pre-fills the payable account, currency, exchange rate (looked up by invoice date) and per-line expense account, learning from prior saves.	Partial	Pre-fill	High Production
invoice-auto-regen Auto-regenerar Facturas	Detects SAT-stamped invoices whose PDF was generated <i>before</i> the stamp (missing the SAT seal, QR and fiscal folio) and flags them for regeneration so the PDF is complete and valid.	No	Triggers native	High (fiscal) Stable (post-fix)
invoice-listing-marker Marcadores de Facturas	Color-codes the invoice list: flags zero-amount invoices, credit notes (negative totals) and un-stamped drafts, so they're caught at a glance during month-end review.	No	Read-only	Medium Production
invoice-default-tab Tab por defecto en Invoices	Opens the Invoices section on the "Packing Slips" tab (Eco's start of the billing flow) instead of the default tab; respects the user's choice once they navigate.	No	Read-only	Low (convenience) Production

D. Purchasing & order reconciliation (customer PO ↔ sales order)

Turning customer POs (PDF / portal exports) into matched, validated sales orders. These use AI only to parse the incoming PO/portal document.

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
portal-importer Importador de Portales	Imports a customer purchasing-portal export (e.g. Hubbell), reconciles each PO against Steelhead (exists / candidate / new), and creates the missing sales orders in batch — resolving part numbers with a learned mapping memory.	No	Read / Write (rev.)	Critical Production

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
po-comparator Validador OC vs OV	Reads the customer PO PDF, finds the matching sales order, and presents a line-by-line comparison (price / qty / partial / missing) so discrepancies are caught before production; can notify the team and re-order lines to match the PO.	No	Read / Write (rev.)	High Production
po-reconciler Reconciliador OV vs PO Schneider	For Schneider QRO: matches the temporary sales orders opened before the real SAP PO arrives, computes a re-assignment plan (which WOs move, which SO is renamed to the SAP number, remainder handling), shows it for approval, then executes it.	No	Read / Write (rev.)	High Validated (needs supervision)
so-line-merge Planned	Bulk-merges line rows within a sales order according to defined criteria (e.g. same part / product / price), consolidating many rows into one in a single pass instead of editing the order line by line. <i>Criteria and scope still being defined.</i>	No	TBD (planned)	High Not yet developed

E. Production, work orders & inventory operations

Shop-floor and inventory operations Steelhead handles one record at a time, done in bulk or in real time.

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
paros-linea Paro de Línea	A floating Andon button to register, time and close line-stop events (maintenance events) in real time from any workboard or WO — capturing area, equipment, reason and photo evidence. Feeds OEE directly.	Partial	Write (rev.)	Critical (OEE) Production
wo-deadline-changer Gestión Masiva de OT	Selects work orders by multi-criteria filters (customer, part, product, process, SO, dates) and applies a new deadline or adds/removes tags in bulk — for mass re-prioritization.	No	Write (rev.)	High Production

Tool (launcher name)	What it automates	Native?	Data access	Importance / Status
wo-mover Mover OTs entre OV's	From a sales order, reassigns its work orders to another sales order of the same customer (existing or new) from a multi-select table, instead of editing each WO header individually.	Partial	Write (rev.)	Medium Production (pending e2e)
inventory-reset † Reinicio de Inventario	Two-phase inventory reset: archives all active batches of the selected inventory types, then creates fresh batches with starting quantities from a CSV — used at period close / audits. Confirmation-gated.	No	Write †	High Validated
warehouse-issue-slip Planned	Bulk generation/registration of warehouse issue slips (“vales de salida de almacén”) when material leaves the warehouse, to tighten internal control over what was issued, for which order, and by whom. <i>Not billing-critical; reduces internal-control friction. Scope still being defined.</i>	No	TBD (planned)	Medium Not yet developed

F. Reports, exploration & utilities

Reporting convenience and project infrastructure.

Tool (launcher name)	What it does	Native?	Data access	Importance / Status
report-regen Regenerar Reportes	A one-click button to regenerate the reports database, with live progress and a shared cooldown timer — exposing a function that already exists but is buried in the UI.	Partial	Triggers native	Low (convenience) Pending validation
report-liberator † Liberador de Reportes	Admin maintenance for reports: bulk-unarchive, move reports out of folders to make them editable, and optionally delete empty folders (the delete step is the only irreversible action, and is confirmation-gated).	No	Write (rev.) †	Low (admin) Validated

Tool (launcher name)	What it does	Native?	Data access	Importance / Status
hash-scanner Explorador Steelhead	Internal infrastructure: passively observes our own session traffic to detect when Steelhead rotates its operation identifiers, so our tools stay in sync (sensitive values are sanitized before logging).	n/a (infra)	Read-only	Internal Production

G. Standalone DevTools (no launcher icon)

Not part of the extension launcher — one-off scripts the technical team runs by hand for recovery/audit. Listed for completeness; mostly read-only.

Tool	What it does	Native?	Data access	Importance / Status
audit-incomplete-pns	Compares the Steelhead catalog against the source load CSV and emits a recovery CSV of any gaps (tags, specs, racks, price, dimensions, process).	No	Read-only	High (QA) Live-validated
process-deep-audit	Audits all processes (structure, cycle times, duplicates) and exports an Excel of findings with suggested actions. (Ships within the <i>process-canon</i> bundle.)	No	Read-only	Medium Live-validated
dual-source-recovery	One-shot reconciliation of original quoting Excels vs the Steelhead report; emits a correction template (handled a specific data-migration incident).	No	Read-only	One-time Consumed
wb-produccion-access	Grants the “WB Producción” workboard to a pasted list of users at once (resolves names to IDs), vs. adding users one by one.	Partial	Write (rev.)	Low (admin) Live-validated
archive-inventory-batch-statuses	Lists inventory-batch statuses, detects which have active batches, and bulk-archives the selected ones — diagnosing the generic error the native UI throws.	Partial	Write (rev.)	Low (admin) Live-validated

At a glance

- **30 tools in the extension launcher + 5 standalone DevTools = 35 in use, across 7 categories** — plus **4 high-priority capabilities we've identified but not yet built** (*engineering-calc*, *tank-autoroute*, *so-line-merge*, *warehouse-issue-slip*).
- **The clear majority address gaps Steelhead has no native equivalent for today.** Where it's marked "Partial," the function exists but is per-record, buried in the UI, or has no bulk path.
- **Data-safety footprint is low by construction.** A meaningful share are read-only or merely pre-fill forms the user saves; the rest write reversibly (archive, not delete) and audited. Only three tools (†: *inventory-reset*, *process-canon apply*, *report-liberator's folder delete*) perform higher-impact operations, and each is gated behind explicit user confirmation.
- **The operationally load-bearing ones are specific:** the billing flow (*cfdi-attacher*, *invoice-autofill*, *invoice-auto-regen*), order intake (*portal-importer*, *po-reconciler*), the part catalog (*bulk-upload*, *process-canon*) and shop-floor OEE (*paros-linea*). Billing in particular is part of why we cannot yet retire the legacy system — so these are where native parity matters most before any transition.

Prepared by Omar Viazcán for Eco Plating — happy to walk Steelhead's engineering team through any tool in detail.